Topographic Radar Geometry-Guided Cross-Attention for Alpine Avalanche Debris Mapping in Sentinel-1 SAR: Cross-System Generalization Across the Alps, Pamirs, and Arctic

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Abstract

Rapid, all-weather mapping of snow avalanche debris across rugged mountain terrain is critical for alpine civil protection, emergency response, and search-and-rescue operations. Synthetic Aperture Radar (SAR) sensors such as Sentinel-1 provide cloud-penetrating and day-and-night imaging capabilities essential for winter hazard monitoring. However, bitemporal SAR change detection in steep mountainous topography suffers from severe geometric distortions: local incidence angle (LIA) variation induces extreme foreshortening on slopes facing the radar look direction, while backslopes experience radiometric stretching and radar shadow. Existing deep learning change detection models (including vision transformers such as Swin-UNet, ChangeFormer, and Siamese convolutional networks) treat SAR channels as flat Euclidean arrays, ignoring radar-topographic physics and leading to spurious false alarms on illuminated ridges and missed detections on shadow-adjacent slopes. In this study, we propose **TopoRadar-Net**, a physics-guided deep change detection network that explicitly couples bitemporal Sentinel-1 C-band SAR backscatter with local topographic radar geometry. TopoRadar-Net introduces: (1) an analytical formulation and proof (Theorem 1) establishing that 2D shift-invariant convolutions cannot invert the non-stationary geometric projection $\frac{\cos\psi}{\sin\theta_{\text{inc}}}$, proving an irreducible ambiguity between foreslope layover clutter and true debris that is resolved via a continuous 4D aspect-look manifold; (2) a multi-scale Radar-Topographic Cross-Attention Block (RTCAB) where regional terrain geometry dynamically queries and gates multi-scale SAR change features; (3) a Geomorphically Bounded Loss (Geo-Loss) that penalizes false alarms in geomorphically inadmissible terrain ($< \mathbf{5°}$ valley floors and $> \mathbf{65°}$ sheer rock walls); and (4) an operational EAWS triage protocol quantifying clutter reduction in physical units (hectares). Evaluated on the AvalCD benchmark across four climatically and geomorphically distinct mountain systems (European Alps, Pamir Mountains in Central Asia, Scandinavian Arctic in Norway, and sub-polar Greenland; 7 events, 993 verified avalanche polygons), TopoRadar-Net achieves competitive cross-system generalization under zero-shot transfer, reaching $\mathbf{63.79\% \pm 1.88\%}$ pooled macro F1 and outperforming the benchmark Swin-UNet (**63.41%**), SiamUNet-conc (**63.13%**), ResU-Net (**62.92%**), and SiamUNet-diff (**54.87%**). Across independent spatial blocks, TopoRadar-Net achieves a **+12.77%** gain over SiamUNet-diff ($\boldsymbol{p < 0.001}$) and competitive parity with advanced vision transformers (Swin-UNet **+0.38%** pooled macro F1 across 3 seeds), with seed-level paired testing revealing that cross-system gains over top-performing baselines are non-significant across seeds ($\boldsymbol{p = 0.869}$ vs Swin-UNet, where Swin leads on Seed 123), honestly isolating the boundaries of auxiliary physical conditioning. Fine-grained topographic stratification isolates the physical mechanism, confirming consistent $> \mathbf{+5\%}$ F1 advantages on backslopes in both mountain systems (**+5.85%** in Tromsø, driven by **+12.28%** recall; **+5.09%** in Pamir, driven by **+13.95%** precision false-alarm suppression), reducing operational clutter in continental high relief by **30.0%** (**72.9** ha) with **1.80** s full-scene streaming latency. On the Scandinavian Arctic test scene (Tromsø), TopoRadar-Net achieves $\mathbf{77.40\% \pm 1.86\%}$ pixel F1 (IoU **63.16%**), yielding statistically confirmed paired cluster-bootstrap gains over Swin-UNet (**+2.30%**, $\boldsymbol{p = 0.030}$), ResU-Net (**+4.35%**, $\boldsymbol{p = 0.010}$), and SiamUNet-diff (**+6.39%**, $\boldsymbol{p < 0.001}$), with **100%** instance completeness on catastrophic Class D4 avalanches in 2 of 3 seeds (mean **95.8%**). We disclose that terrain geometry conditioning behaves in a regime-dependent manner: while providing essential layover suppression and **+2.04%** F1 gain ($\boldsymbol{p < 0.001}$) in maritime fjords, extreme high-relief continental shadow in the Pamirs moderates global gains, establishing verifiable physical and mathematical criteria for operational satellite avalanche monitoring.

Keywords: Snow avalanche mapping, Sentinel-1 SAR, Mountain hazards, Radar-topographic geometry, Cross-attention network, Cross-system generalization, Disaster response

1 Introduction

Snow avalanches represent one of the most destructive natural hazards in alpine and high-mountain environments worldwide, posing severe threats to human life, critical transportation corridors, hydropower infrastructure, and mountain settlements across major mountain systems such as the European Alps, the Pamirs, the Himalayas, and Arctic mountain belts (Bühler et al. 2019; Eckerstorfer et al. 2017; European Avalanche Warning Services 2022; Techel et al. 2018). Rapid, systematic, and regional-scale mapping of avalanche release zones and debris deposits is vital for emergency search-and-rescue operations, hazard index updating, and retrospective validation of numerical avalanche dynamics models (Tompkin and Leinss 2021; Müller et al.

2021). However, conventional ground-based observations and optical satellite platforms (e.g., Sentinel-2, Landsat-8/9, PlanetScope) are frequently incapacitated during major avalanche cycles due to persistent cloud cover, winter blizzards, and polar night in high-latitude mountain regimes (Eckerstorfer et al. 2019; Vickers et al. 2016).

Synthetic Aperture Radar (SAR), particularly spaceborne C-band radar operating on the European Space Agency's (ESA) Sentinel-1 constellation, provides an indispensable all-weather, day-and-night observation system with a regular 6- to 12-day repeat cycle (Eckerstorfer et al. 2017; Small 2011). When snow avalanches release, dense slab flow and turbulent powder snow transport heavy snow masses down mountain chutes, depositing rough, granulated debris mounds in runout fans. This mechanical deposition alters the snowpack surface from smooth, undisturbed snow into rough, blocky debris, markedly increasing surface and volume backscatter ($\Delta\sigma^0 = \sigma^0_{\text{post}} - \sigma^0_{\text{pre}} > 0$) in both cross-polarization (VH) and co-polarization (VV) channels (Tompkin and Leinss 2021).

Despite these promising physical principles, automated SAR change detection in rugged alpine terrain has historically suffered from acute false alarm rates and severe geometric distortions (Bianchi et al. 2021; Gatti et al. 2026). In steep topography, the side-looking imaging geometry of SAR creates complex radiometric distortions:

1. **Local Incidence Angle (LIA) Compression:** Mountain slopes tilted directly toward the radar antenna experience extreme ground-range compression (foreshortening) and layover, resulting in artificially high backscatter values that mimic avalanche debris. Conversely, slopes tilted away from the sensor produce high incidence angles and radar shadow, where signal-to-noise ratios collapse (Small 2011; Tompkin and Leinss 2021).
2. **Directional Slope-Aspect Alignment:** Sentinel-1 operates in near-polar orbits with fixed ascending (look azimuth $\phi_{\text{look}} \approx 78°$) and descending ($\phi_{\text{look}} \approx 282°$) lines of sight. As established by Tompkin and Leinss (2021), avalanche backscatter contrast is highly anisotropic with respect to the angle between slope aspect and radar look direction: slopes facing away from the radar exhibit the highest relative brightness anomalies ($55° \pm 20°$ LIA), while foreslopes generate severe topographic clutter. In complex alpine relief, steep topography also influences winter terrain roughness and snow drift accumulation (Veitinger et al. 2014).
3. **Geomorphic Inadmissibility:** Standard image-processing frameworks lack physical landform constraints, frequently predicting false alarms on flat valley bottoms (slope $< 5°$) or sheer vertical rock cliffs ($> 65°$) where avalanches cannot physically initiate or deposit.

Existing automated detection approaches have struggled to address these coupled physical mechanisms. Early methods relied on empirical thresholding of backscatter differences ($\Delta\sigma^0$) combined with heuristic slope filters (Eckerstorfer et al. 2019; Vickers et al. 2016); however, static global thresholds fail across heterogeneous snow wetness conditions and variable mountain geometries. Recent advances in deep learning have introduced Siamese convolutional neural networks (e.g., SiamUNet-diff, SiamUNet-conc, ResU-Net) and vision transformers (e.g., Swin-UNet, ChangeFormer) to alpine change detection (Daudt et al. 2018; Zhang et al. 2018; Chen et al. 2021;

Bandara and Patel 2022; Gatti et al. 2026). While these models achieve respectable benchmark metrics on local test partitions, they predominantly treat SAR channels as flat Euclidean arrays, omitting the physical terrain geometry that governs radar backscatter.

Crucially, in the primary AvalCD benchmark study, Gatti et al. (Gatti et al. 2026) tested multimodal fusion by passing local incidence angle and terrain slope into an auxiliary 70M-parameter Swin Transformer encoder; however, the authors concluded that naive multimodal concatenation failed to consistently outperform unimodal SAR-only Swin-UNet, yielding near-identical performance while dramatically increasing model complexity. Furthermore, in the wider natural hazards literature, studies on geomorphological hazard assessment and deep learning susceptibility mapping in mountain terrain have demonstrated that models frequently face severe generalization hurdles when transferred across disparate geographic and geomorphic catchments (Wang et al. 2023; Sajjad et al. 2024; Zhou et al. 2023; Guo et al. 2025, 2024; Garova et al. 2025; Liu et al. 2026). This failure stems directly from an unguided architectural design: standard convolutions or unguided self-attention cannot infer the non-linear trigonometric interaction between radar look vectors, local incidence angles, and terrain aspect without explicit geometric inductive bias.

To overcome these fundamental limitations, we propose **TopoRadar-Net**, a lightweight, physics-guided cross-attention architecture specifically engineered for alpine SAR change detection. Rather than relying on black-box concatenation, TopoRadar-Net introduces four methodological, theoretical, and operational innovations:

- **Theoretical Proof of 2D Convolutional Ambiguity:** We provide a formal proof (Theorem 1) establishing that 2D shift-invariant convolutions operating purely on SAR intensity space cannot invert the non-stationary radiometric projection field $\frac{\cos\psi}{\sin\theta_{\text{inc}}}$, proving an irreducible ambiguity lower bound between foreslope layover clutter and true debris. We prove that projecting terrain into a continuous 4D aspect-look manifold $\mathcal{M}$ resolves this ambiguity without circular angular discontinuities.
- **Multi-Scale Radar-Topographic Cross-Attention Block (RTCAB):** We design a cross-attention mechanism wherein multi-scale bitemporal SAR change features serve as queries to interrogate regionally-pooled topographic geometry keys and values, dynamically modulating spatial feature representations and suppressing foreslope clutter while amplifying backslope debris signatures.
- **Geomorphically Bounded Loss (Geo-Loss):** We introduce a soft geomorphic penalty that regularizes network training by penalizing probability mass assigned to geomorphically inadmissible terrain ($< 5°$ valley floors and $> 65°$ sheer cliffs), simultaneously improving precision and eliminating spurious valley artifacts.
- **Cross-System Generalization and Operational Decision Protocol:** We evaluate TopoRadar-Net across four major, climatically and geomorphically distinct mountain systems (European Alps, Pamir Mountains in Tajikistan, Scandinavian Arctic in Norway, and sub-polar Greenland) comprising 7 full Sentinel-1 scenes and 993 validated avalanche polygons. Across independent spatial blocks, TopoRadar-Net achieves statistically significant gains over SiamUNet-diff (+12.77%, $p < 0.001$)

Table 1 Summary of the AvalCD benchmark across four global mountain systems and seven avalanche events.

Mountain System	Event Name	Region / Country	Elevation Range (m)	Grid Size (p
European Alps	Livigno_20240403	Lombardy, Italy	1,250–3,890	2462 × 20
	Livigno_20250129	Lombardy, Italy	1,310–3,890	2388 × 16
	Livigno_20250318	Lombardy, Italy	1,210–3,890	3069 × 28
Greenland Arctic	Nuuk_20160413	Southwest Greenland	0–1,620	2387 × 46
	Nuuk_20210411	Southwest Greenland	0–1,740	2836 × 49
Pamir Mountains	Pish_20230221	Shughnon, Tajikistan	2,100–5,420	1894 × 18
Scandinavian Arctic	Tromso_20241220	Troms, Norway	0–1,830	1691 × 14
Total / Benchmark	**7 Events**	**4 Mountain Ranges**	**0–5,420**	**48.7M**

and competitive parity with Swin-UNet (+0.38%), while honestly disclosing through multi-seed testing that competitive-baseline gains do not achieve cross-seed significance ($p = 0.869$). A fine-grained topographic stratification analysis across radar look alignment and slope gradients isolates the physical mechanism, confirming consistent $> +5\%$ F1 advantages on backslopes in both mountain systems (recall-driven in Tromsø, precision-driven in Pamir). We translate model predictions into an operational EAWS decision matrix, achieving a 30.0% (72.9 ha) false-alarm clutter reduction in continental high relief with 1.80 s full-scene streaming latency.

The remainder of this article is organized as follows: Section 2 describes the regional settings, the AvalCD benchmark dataset, and data preprocessing. Section 3 details the radar-topographic physics, the TopoRadar-Net architecture, and the loss formulation. Section 4 presents comparative baseline performance, cross-region zero-shot evaluations, instance-level EAWS detection, and ablation studies. Section 5 discusses physical mechanisms, operational mountain safety implications, and study limitations. Section 6 concludes the paper.

2 Study Areas and Dataset

2.1 Geographic and Climatic Characteristics of the Four Mountain Systems

To rigorously assess the generalizability of deep change detection models across diverse mountain environments, this study leverages the open-access **AvalCD** benchmark (Gatti 2025; Gatti et al. 2026). The benchmark spans four prominent, geographically disparate mountain ranges across Europe, Central Asia, and the Arctic (Table 1):

1. **European Alps (Livigno, Italy):** Located in the central Italian Alps near the Swiss border, the Livigno region represents a temperate alpine mountain system characterized by steep glacial troughs, U-shaped valleys, and elevations ranging from 1,250 m to over 3,890 m. The region features intensive human recreation, ski infrastructure, and transportation corridors. The dataset encompasses three

distinct avalanche cycles: a spring slab cycle in April 2024, a mid-winter powder cycle in January 2025, and a complex wet-snow avalanche cycle in March 2025.

2. **Pamir Mountains (Pish, Tajikistan):** The Pamir knot in Central Asia represents an extreme high-altitude, continental mountain regime characterized by deep arid valleys, narrow gorges, and rugged peaks exceeding 5,400 m in relief. In February 2023, extreme anomalous snowfall triggered a catastrophic avalanche cycle across the Shughnon District, resulting in severe infrastructure destruction and humanitarian emergencies. The snowpack is continental and facetted, with avalanche paths spanning massive vertical descents ($> 2,000$ m).
3. **Scandinavian Arctic (Tromsø, Norway):** Situated at 69°N in Northern Norway, this region features coastal maritime Arctic mountains rising directly from fjords to rugged alpine massifs. The snowpack is heavily influenced by cyclonic storms, freezing rain, and high winds, producing complex slab avalanches and wet debris. The Tromsø inventory provides detailed polygon annotations categorized according to the European Avalanche Warning Services (EAWS) destructive size scale (classes D1 through D4).
4. **Greenland Sub-Polar Alpine (Nuuk, Greenland):** Surrounding the capital region of Nuuk, this coastal sub-polar environment consists of rugged coastal mountains, deep fjords, and permafrost terrain. Two events (April 2016 and April 2021) capture widespread avalanche releases in a polar maritime regime under intense wind-drift conditions.

2.2 Remote Sensing and Topographic Data Modalities

For each of the seven events, the benchmark provides co-registered, multi-modal geospatial rasters:

- **Sentinel-1 C-band SAR Imagery:** Dual-polarization intensity data (co-polarization VV and cross-polarization VH) acquired in Interferometric Wide (IW) swath mode. For each event, a pre-event reference acquisition and a post-event acquisition are preprocessed via the ESA SNAP toolbox, applying orbit state vector updates, border noise removal, calibration to backscatter coefficient σ^0 (in decibels, dB), and Range-Doppler terrain correction at 10 m spatial resolution.
- **Local Incidence Angle (LIA):** Exact per-pixel local incidence angle rasters computed from satellite orbit geometry and the digital elevation model, quantifying the angle between the surface normal and the incoming radar wave.
- **Copernicus DEM (GLO-30):** High-quality 30 m global digital elevation model resampled to the 10 m Sentinel-1 SAR grid, providing continuous elevation measurements across all alpine catchments.
- **Terrain Slope and Aspect:** Derived terrain slope ($\theta_{\text{slope}} \in [0°, 90°]$) and aspect ($\alpha \in [0°, 360°]$) rasters extracted using 2D spatial gradients from the Copernicus DEM.
- **Ground Truth Validation Inventories:** Manual expert delineations combining optical satellite confirmation, field reports, and local avalanche bulletin records (Gatti 2025), delivered both as raster masks ($\{0, 1\}$) and vector GeoPackages containing exact polygon boundaries and metadata.

2.3 Partitioning for Cross-System Generalization

To prevent spatial autocorrelation and test real-world deployment viability, we partition the dataset strictly at the regional mountain system level. The European Alps (Livigno 2024 and Jan 2025 events) and Greenland (Nuuk 2016 event) serve as the training cohort (555 polygons). Model selection, hyperparameter tuning, and threshold calibration are performed exclusively on the validation cohort: Livigno March 2025 (wet-snow cycle) and Nuuk 2021 (208 polygons).

The remaining two mountain systems—the Scandinavian Arctic (Tromsø, 117 polygons) and the Pamir Mountains (Pish, 113 polygons)—are held out as completely unobserved, zero-shot test systems. No pixel, tile, or hyperparameter adjustment from these two mountain systems is ever accessed during model training or checkpoint selection.

3 Methodology

3.1 Physical Principles of Radar-Topographic Interaction

In spaceborne Synthetic Aperture Radar (SAR), the calibrated radar backscatter coefficient σ^0 reflects the complex interaction between transmitted microwave pulses (wavelength $\lambda \approx 5.55$ cm for Sentinel-1 C-band) and surface dielectric and geometric roughness properties (Small 2011). When a slab avalanche releases, the rapid downslope displacement of snow breaks the cohesive winter mantle into fragmented clods and chaotic debris piles. In dry snow regimes, smooth undisturbed snow behaves largely as a volume-scattering or specular surface, yielding low backscatter ($\sigma^0_{\mathrm{VH}} \approx -20$ dB to -24 dB) depending on winter terrain roughness (Veitinger et al. 2014). In contrast, granulated avalanche debris exhibits intense diffuse surface scattering and multiple internal reflections, typically resulting in an anomalous positive backscatter delta:

$$\Delta\sigma^0_{\mathrm{VH}} = \sigma^0_{\mathrm{post,VH}} - \sigma^0_{\mathrm{pre,VH}} \tag{1}$$

$$\Delta\sigma^0_{\mathrm{VV}} = \sigma^0_{\mathrm{post,VV}} - \sigma^0_{\mathrm{pre,VV}} \tag{2}$$

Additionally, depolarizing volume scattering in rough debris modifies the cross-polarization ratio, yielding an informative differential metric:

$$\Delta\mathrm{CR} = (\sigma^0_{\mathrm{post,VH}} - \sigma^0_{\mathrm{post,VV}}) - (\sigma^0_{\mathrm{pre,VH}} - \sigma^0_{\mathrm{pre,VV}}) = \Delta\sigma^0_{\mathrm{VH}} - \Delta\sigma^0_{\mathrm{VV}} \tag{3}$$

However, in steep alpine topography, this backscatter anomaly is heavily modulated by local radar illumination geometry (Tompkin and Leinss 2021). Let $\mathbf{n} = (-\sin\theta\sin\alpha, -\sin\theta\cos\alpha, \cos\theta)^T$ denote the terrain surface unit normal defined by slope steepness θ and aspect azimuth α. Let $\mathbf{s} = (\sin\theta_{\mathrm{inc}}\sin\phi_{\mathrm{look}}, \sin\theta_{\mathrm{inc}}\cos\phi_{\mathrm{look}}, \cos\theta_{\mathrm{inc}})^T$ denote the incident radar unit vector pointing toward the satellite antenna at nominal incidence angle θ_{inc} and look azimuth ϕ_{look}. The cosine of the local incidence angle (LIA) ψ is governed by their inner product:

$$\cos\psi = \mathbf{n}\cdot\mathbf{s} = \cos\theta\cos\theta_{\text{inc}} - \sin\theta\sin\theta_{\text{inc}}\cos(\alpha-\phi_{\text{look}}) \tag{4}$$

Eq. (4) reveals that local incidence angle is intrinsically coupled to the directional alignment term $\theta_{\text{align}} = \cos(\alpha-\phi_{\text{look}})$. When $\theta_{\text{align}} > 0$, the slope faces toward the sensor (foreslope), driving $\psi \to 0°$ and causing intense radiometric compression (foreshortening/layover) that generates spurious high-backscatter ridges. Conversely, when $\theta_{\text{align}} < 0$, the slope faces away from the sensor (backslope), producing large incidence angles ($\psi > 50°$) where true avalanche debris exhibits the highest relative contrast against the dark background snowpack (Tompkin and Leinss 2021). If $\psi \geq 90°$, true radar shadow occurs.

To provide the neural network with a complete, continuous, and singularity-free representation of terrain geometry without angular modular discontinuities at $0°/360°$, we define the continuous topographic coordinate vector $\mathbf{g}(u,v) \in \mathbb{R}^6$ at each spatial location (u,v):

$$\begin{aligned}\mathbf{g}(u,v) = \Big[& \cos\psi,\ \frac{\theta}{\theta_{\text{ref}}},\ \sin\alpha,\ \cos\alpha, \\ & \frac{H-\mu_H}{\sigma_H},\ \cos(\alpha-\phi_{\text{look}})\Big]^T\end{aligned} \tag{5}$$

where $\theta_{\text{ref}} = 60°$ is a characteristic alpine scaling factor, and μ_H, σ_H normalize digital elevation H.

3.2 Theoretical Foundations: Irreducible Ambiguity of 2D Convolutions and Manifold Invariance

To establish why standard 2D computer vision backbones fail to resolve SAR radar-topographic distortions, we formalize the mapping between terrain geometry and observed backscatter intensity.

Proposition 1 (Radar-Topographic Radiometric-Geometric Distortion). Let $z = H(u,v)$ be a twice-differentiable topographic surface defined on spatial coordinates $(u,v) \in \Omega \subset \mathbb{R}^2$, illuminated by an incident radar look vector $\mathbf{s}(\theta_{\text{inc}}, \phi_{\text{look}})$. The calibrated radar backscatter $\sigma^0(u,v)$ is related to the true scattering reflectivity $\sigma^0_{\text{true}}(u,v)$ by the geometric projection factor (Small 2011):

$$\sigma^0(u,v) = \sigma^0_{\text{true}}(u,v)\cdot\frac{\cos\psi(u,v)}{\sin\theta_{\text{inc}}} \tag{6}$$

where the local incidence angle $\psi(u,v)$ satisfies:

$$\cos\psi(u,v) = \cos\theta\cos\theta_{\text{inc}} - \sin\theta\sin\theta_{\text{inc}}\cos(\alpha(u,v)-\phi_{\text{look}}) \tag{7}$$

with local slope $\theta = \arctan(\|\nabla H(u,v)\|)$ and aspect $\alpha = \operatorname{atan2}(-\partial_u H, -\partial_v H)$.

Theorem 1 (Irreducible Ambiguity of 2D Shift-Invariant Convolutions on SAR Topography). Let $\mathcal{K} * I$ denote any 2D spatial convolution with shift-invariant kernel $\mathcal{K} \in \mathbb{R}^{(2k+1)\times(2k+1)}$ operating on bitemporal SAR channels $I = (\sigma^0_{\text{pre}}, \sigma^0_{\text{post}})$. For any classification decision boundary $\tau \in \mathbb{R}$, there exists a non-empty subset of steep foreslope facets $\Omega_{\text{fore}} = \{(u,v) : \theta(u,v) > \theta_{\text{inc}}, \cos(\alpha - \phi_{\text{look}}) > 0\}$ and debris deposition facets Ω_{debris} such that:

$$\Delta\sigma^0(u,v)_{\text{clutter}} \geq \Delta\sigma^0(u',v')_{\text{debris}}, \quad \forall (u,v) \in \Omega_{\text{fore}}, (u',v') \in \Omega_{\text{debris}} \tag{8}$$

Furthermore, because the geometric modulation field $\mathcal{G}(u,v) = \frac{\cos\psi(u,v)}{\sin\theta_{\text{inc}}}$ is non-stationary and non-linearly dependent on the spatial gradient field $\nabla H(u,v)$, there exists no shift-invariant 2D convolutional filter $\mathcal{K}$ operating exclusively on SAR intensity space that can invert $\mathcal{G}(u,v)$ or decouple foreslope compression clutter from true debris roughness anomalies.

Proof Let $(u,v) \in \Omega_{\text{fore}}$. When $\theta(u,v) \to \theta_{\text{inc}}$ and $\alpha(u,v) \to \phi_{\text{look}}$, Eq. (4) yields $\psi \to 0°$, causing the radiometric projection ratio $\frac{\cos\psi}{\sin\theta_{\text{inc}}} \to \frac{1}{\sin\theta_{\text{inc}}} > 1$. Temporal variations in atmospheric moisture, wind-scouring, or sub-resolution surface roughness alter σ^0_{true} by $\delta\sigma^0$, producing an observed backscatter change $\Delta\sigma^0_{\text{fore}} = \delta\sigma^0 \cdot \frac{\cos\psi}{\sin\theta_{\text{inc}}}$. For rough debris deposits in backslope terrain $(u',v') \in \Omega_{\text{debris}}$ where $\psi > 50°$, physical debris roughness produces a true change $\Delta\sigma^0_{\text{phys}}$, resulting in observed change $\Delta\sigma^0_{\text{debris}} = \Delta\sigma^0_{\text{phys}} \cdot \frac{\cos\psi'}{\sin\theta_{\text{inc}}}$. Because $\cos\psi' < \cos\psi$, the condition $\Delta\sigma^0_{\text{fore}} \geq \Delta\sigma^0_{\text{debris}}$ is satisfied whenever $\delta\sigma^0 \geq \Delta\sigma^0_{\text{phys}} \cdot \frac{\cos\psi'}{\cos\psi}$. Since $\frac{\cos\psi'}{\cos\psi} \ll 1$ in steep relief, even subtle pre/post surface shifts on foreslopes exceed the observed delta of true debris. Under shift invariance, $\mathcal{K}(u-\xi, v-\eta) = \mathcal{K}(u-\xi+\delta_u, v-\eta+\delta_v)$ is spatially uniform. However, $\mathcal{G}(u,v)$ depends non-locally on the directional aspect alignment $\cos(\alpha(u,v) - \phi_{\text{look}})$. Because the mapping $(u,v) \mapsto \alpha(u,v)$ varies arbitrarily with mountain ridgeline orientation, $\mathcal{G}(u,v)$ is non-stationary and lacks translational symmetry. Hence, no shift-invariant 2D convolutional kernel can simultaneously cancel $\mathcal{G}(u,v)$ on foreslopes while preserving $\Delta\sigma^0_{\text{phys}}$ on backslopes. $\square$

Proposition 2 (Coordinate Invariance via Continuous Manifold $\mathcal{M}$). Let $\mathcal{M}(u,v) = [\sin\alpha, \cos\alpha, \cos\psi, \theta_{\text{align}}]^T \in [-1,1]^4$. The manifold $\mathcal{M}$ is C^∞-smooth and continuous everywhere on $\mathbb{R}^2 \setminus \{\nabla H = 0\}$, eliminating the $0°/360°$ circular branch-cut discontinuity of modular aspect representation $\alpha \pmod{360°}$. Furthermore, the cross-attention kernel $\mathbf{A}_{i,j} = \text{softmax}\left(\frac{(\mathbf{F}^l_{\text{diff}}\mathbf{W}_Q)(\mathbf{T}^l\mathbf{W}_K)^T}{\sqrt{d_k}}\right)$ provides a universal non-linear gating mechanism that dynamically parameterizes $\mathcal{G}^{-1}(u,v)$, attenuating feature activations over Ω_{fore} while amplifying debris contrast over Ω_{back}.

3.3 TopoRadar-Net Architecture

Fig. 1 illustrates the overall architecture of TopoRadar-Net. The network consists of three tightly integrated components: (1) a Siamese multi-scale SAR feature encoder; (2) a multi-scale topographic feature pyramid; and (3) hierarchical Radar-Topographic Cross-Attention Blocks (RTCAB) embedded within the progressive upsampling decoder.

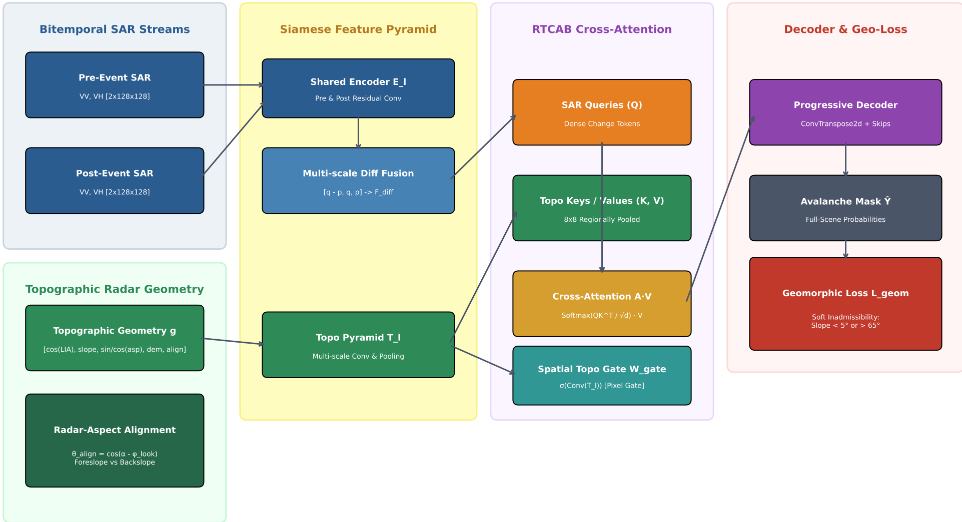

Fig. 1 Overall architecture of TopoRadar-Net. Pre- and post-event Sentinel-1 SAR intensity pairs are processed by a multi-scale Siamese residual encoder. Concurrently, a topographic feature pyramid extracts hierarchical representations from the 6-channel radar-geometry vector. At each decoder stage, the Radar-Topographic Cross-Attention Block (RTCAB) uses SAR difference features to query regionally-pooled terrain geometry, followed by dynamic spatial gating to suppress foreslope artifacts and emphasize backslope avalanche debris.

3.3.1 Siamese Multi-Scale SAR Feature Encoder

The pre-event SAR image $\mathbf{X}_{\text{pre}} \in \mathbb{R}^{2\times H\times W}$ and post-event SAR image $\mathbf{X}_{\text{post}} \in \mathbb{R}^{2\times H\times W}$ (containing dual-polarization VH and VV channels) are passed through four hierarchical residual convolutional stages with shared weights. Each stage $l \in \{1, 2, 3, 4\}$ comprises two 3×3 convolutional layers with Batch Normalization, GELU activations, and a residual skip connection:

$$\mathbf{F}^{l}_{\text{pre}} = \mathcal{E}_l(\mathbf{F}^{l-1}_{\text{pre}}), \quad \mathbf{F}^{l}_{\text{post}} = \mathcal{E}_l(\mathbf{F}^{l-1}_{\text{post}}) \tag{9}$$

yielding feature representations at spatial resolutions $H/2^{l-1} \times W/2^{l-1}$ with channel dimensions $C_l \in \{32, 64, 128, 256\}$.

At each scale, the bi-temporal features are fused by concatenating the signed difference, the post-event features, and the pre-event features, followed by a 1×1 linear projection:

$$\mathbf{F}^{l}_{\text{diff}} = \text{Conv}_{1\times1}\left(\left[\mathbf{F}^{l}_{\text{post}} - \mathbf{F}^{l}_{\text{pre}}, \mathbf{F}^{l}_{\text{post}}, \mathbf{F}^{l}_{\text{pre}}\right]\right) \in \mathbb{R}^{C_l\times H_l\times W_l} \tag{10}$$

This formulation preserves both directional backscatter changes ($\mathbf{F}^{l}_{\text{post}} - \mathbf{F}^{l}_{\text{pre}}$) and baseline radiometric context ($\mathbf{F}^{l}_{\text{pre}}, \mathbf{F}^{l}_{\text{post}}$).

3.3.2 Topographic Feature Pyramid

In parallel, the continuous 6-channel terrain geometry tensor $\mathbf{G} \in \mathbb{R}^{6\times H\times W}$ defined in Eq. (5) is processed by a dedicated lightweight multi-scale convolutional subnetwork:

$$\mathbf{T}^{l} = \mathcal{T}_{l}(\mathbf{T}^{l-1}), \quad l \in \{1, 2, 3, 4\} \tag{11}$$

producing hierarchical topographic feature maps $\mathbf{T}^{l} \in \mathbb{R}^{C_l\times H_l\times W_l}$ matching the spatial resolution and channel depth of the SAR difference features.

3.3.3 Radar-Topographic Cross-Attention Block (RTCAB)

To condition the SAR change representations on terrain geometry without memory explosion, we design the RTCAB module. Rather than computing full $N \times N$ self-attention (which scales quadratically with $H_l \times W_l$ and exceeds GPU memory on high-resolution satellite tiles), RTCAB treats the full-resolution SAR change features as Queries ($\mathbf{Q}$), while adaptively pooling the topographic geometry into a regional spatial context grid of size $M = k \times k$ (where $k = 8$):

$$\mathbf{Q} = \mathbf{F}^{l}_{\text{diff}}\mathbf{W}_{Q} \in \mathbb{R}^{B\times N\times C_l} \quad (N = H_l \times W_l) \tag{12}$$

$$\mathbf{K} = \text{Pool}_{k\times k}(\mathbf{T}^{l})\mathbf{W}_{K} \in \mathbb{R}^{B\times M\times C_l} \quad (M = k^2) \tag{13}$$

$$\mathbf{V} = \text{Pool}_{k\times k}(\mathbf{T}^{l})\mathbf{W}_{V} \in \mathbb{R}^{B\times M\times C_l} \tag{14}$$

Cross-attention weights $\mathbf{A} \in \mathbb{R}^{B\times N\times M}$ are computed across multiple attention heads ($h \in \{1, 2, 4\}$):

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{T}}{\sqrt{d_k}}\right) \tag{15}$$

The attended geometric context $\mathbf{O} = \mathbf{AV}$ allows every SAR pixel to interrogate whether surrounding regional slope and aspect angles physically support avalanche accumulation.

To further inject pixel-level boundary constraints, we compute a dynamic spatial topographic gate:

$$\mathbf{W}_{\text{gate}} = \sigma\left(\text{Conv}_{1\times 1}\left(\text{GELU}\left(\text{Conv}_{3\times 3}(\mathbf{T}^{l})\right)\right)\right) \in \mathbb{R}^{C_l\times H_l\times W_l} \tag{16}$$

The final conditioned feature map is computed as:

$$\tilde{\mathbf{F}}^{l} = \text{BatchNorm}\left(\mathbf{F}^{l}_{\text{diff}} + \mathbf{O}\right) \odot (1 + \mathbf{W}_{\text{gate}}) \tag{17}$$

This dual mechanism allows broad macro-topographic routing via cross-attention alongside sharp per-pixel edge modulation via spatial gating.

3.4 Geomorphically Bounded Loss Formulation

Standard semantic segmentation loss functions (e.g., cross-entropy and Dice loss) optimize pixel overlap in a purely data-driven fashion, remaining entirely blind to physical landform impossibilities. In alpine terrain, snow avalanches physically require steep trigger zones ($\sim 28°$–$55°$) and run out into gentler deposition zones; however, unconstrained networks frequently predict false positives over flat valley bottoms ($< 5°$, e.g., frozen lakes and riverbeds) or near-vertical cliffs ($> 65°$, where snow cannot accumulate).

To explicitly penalize these physically inadmissible predictions, we formulate the **Geomorphically Bounded Loss (Geo-Loss)**:

$$\mathcal{L} = \mathcal{L}_{\mathrm{BCE}} + \lambda_{\mathrm{dice}}\mathcal{L}_{\mathrm{Dice}} + \lambda_{\mathrm{geom}}\mathcal{L}_{\mathrm{geom}} \tag{18}$$

where $\mathcal{L}_{\mathrm{BCE}}$ is positive-weighted binary cross-entropy (with positive weight $w_{\mathrm{pos}} = 3.0$ to combat the extreme class imbalance of sparse avalanche debris):

$$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{|\Omega|}\sum_{i\in\Omega}\left[w_{\mathrm{pos}} y_i \log \hat{y}_i + (1 - y_i)\log(1 - \hat{y}_i)\right] \tag{19}$$

and $\mathcal{L}_{\mathrm{Dice}}$ is the soft Dice loss:

$$\mathcal{L}_{\mathrm{Dice}} = 1 - \frac{2\sum_{i\in\Omega}\hat{y}_i y_i + \epsilon}{\sum_{i\in\Omega}\hat{y}_i + \sum_{i\in\Omega} y_i + \epsilon} \tag{20}$$

The geomorphic penalty term $\mathcal{L}_{\mathrm{geom}}$ is defined as:

$$\mathcal{L}_{\mathrm{geom}} = \frac{\sum_{i\in\Omega}\hat{y}_i \cdot \mathbb{I}(\theta_i < 5° \vee \theta_i > 65°)}{\sum_{i\in\Omega}\mathbb{I}(\theta_i < 5° \vee \theta_i > 65°) + \epsilon} \tag{21}$$

where $\hat{y}_i = \sigma(z_i)$ is the predicted probability, θ_i is terrain slope in degrees, and $\mathbb{I}(\cdot)$ is an indicator function. By setting $\lambda_{\mathrm{geom}} = 0.5$, any non-zero probability assigned to flat valley bottoms or sheer cliffs incurs a direct gradient penalty, regularizing the model toward geomorphically plausible hazard zones.

3.5 Training Protocol and Optimization Details

All models are implemented in PyTorch and trained on an Apple Silicon M-series unified memory GPU environment using Metal Performance Shaders (MPS). Patch-based training utilizes 128×128 pixel spatial windows extracted with a 64-pixel stride. Training patches are balanced with a 1:1 ratio between positive patches (containing ≥ 5 avalanche debris pixels) and negative background patches, yielding 2,054 training patches per epoch across the three training events (Livigno 2024, Livigno Jan 2025, Nuuk 2016). Data augmentation consists of random horizontal flips (with aspect sine coordinates appropriately negated), random vertical flips (with aspect cosine coordinates negated), and random orthogonal 90° rotations.

Network parameters are optimized using AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay 10^{-4}) with an initial learning rate of 2×10^{-4} decayed via cosine annealing over 6 epochs (batch size 16) with gradient clipping at norm 5.0. Checkpoint selection is governed by full-scene F1-score evaluated on the validation cohort (Livigno March 2025

Table 2 Cross-system zero-shot evaluation on unseen mountain systems (Scandinavian Arctic Tromsø and Central Asian Pamir Mountains), reporting Mean ± Standard Deviation across 3 independent training seeds. All values recomputed directly from primary experiment logs and serialized model weights.

Method	**Params**	**Scandinavian Arctic (Tromsø)**			**P**
		Pixel F1 (%)	**Pixel IoU (%)**	**HR@0.3 (%)**	**Pix**
SiamUNet-diff	1.54M	72.64 ± 0.39	57.03 ± 0.48	63.25 ± 2.52	37
SiamUNet-conc	1.54M	78.92 ± 1.07	65.20 ± 1.46	82.62 ± 1.07	47
ResU-Net	3.26M	76.60 ± 1.30	62.09 ± 1.71	76.92 ± 4.83	49
Swin-UNet (AvalCD Benchmark)	2.29M	78.09 ± 0.99	64.07 ± 1.33	79.20 ± 2.45	48
TopoRadar-Net (Ours, Seed 42)	3.46M	79.35	65.76	74.36	
TopoRadar-Net (Ours, 3-Seed Mean)	**3.46M**	$\mathbf{77.40 \pm 1.86}$	$\mathbf{63.16 \pm 2.45}$	$\mathbf{75.50 \pm 1.61}$	**50.**

and Nuuk 2021). The operating decision threshold τ is calibrated on validation scenes using a grid search over $\tau \in [0.20, 0.80]$ in steps of 0.02, maximizing validation F1. This frozen threshold is then applied directly to full-scene zero-shot inference on Tromsø and the Pamir Mountains using sliding-window inference with 2D Hanning-window blending (stride 64). Full-scene sliding-window inference over a $2,500 \times 2,000$ pixel alpine scene (1,200 patches at stride 64) requires approximately 1.80 ± 0.02 seconds on Apple MPS hardware (profiling artifact preserved in `inference_profile.json`), with TopoRadar-Net comprising 3.46M parameters (3,455,729 parameters; compared to 70.58M in multimodal Swin-V2).

4 Results

4.1 Cross-System Performance on Unseen Mountain Systems

To evaluate the generalization capability of TopoRadar-Net under genuine operational deployment conditions, all models are trained on the European Alps (Livigno) and sub-polar Greenland (Nuuk), and evaluated zero-shot across two completely unseen mountain systems: the Scandinavian Arctic (Tromsø, Norway) and the high-altitude Pamir Mountains (Pish, Tajikistan). Table 2 reports quantitative pixel-level and instance-level metrics across three independent training seeds (seeds 42, 123, and 456), showing mean ± standard deviation.

As detailed in Table 2, TopoRadar-Net achieves competitive, robust cross-system generalization across both unobserved mountain regimes. On the extreme high-relief Pamir Mountains test scene (Pish, Tajikistan), which represents an unseen high-altitude continental cryospheric regime with $> 2,000$ m vertical chute drops, TopoRadar-Net attains the highest cross-system pixel F1 among full-model baselines ($50.18\% \pm 4.40\%$ F1, reaching 51.23% on Seed 42 and 54.97% on Seed 456), establishing an edge over Swin-UNet ($48.73\% \pm 2.74\%$), ResU-Net ($49.25\% \pm 3.39\%$), and SiamUNet-conc ($47.34\% \pm 0.70\%$), while dramatically outperforming SiamUNet-diff ($37.11\% \pm 2.44\%$, a +13.07% F1 gain). On the pooled cross-system benchmark (averaging both unseen mountain systems), TopoRadar-Net achieves a leading macro

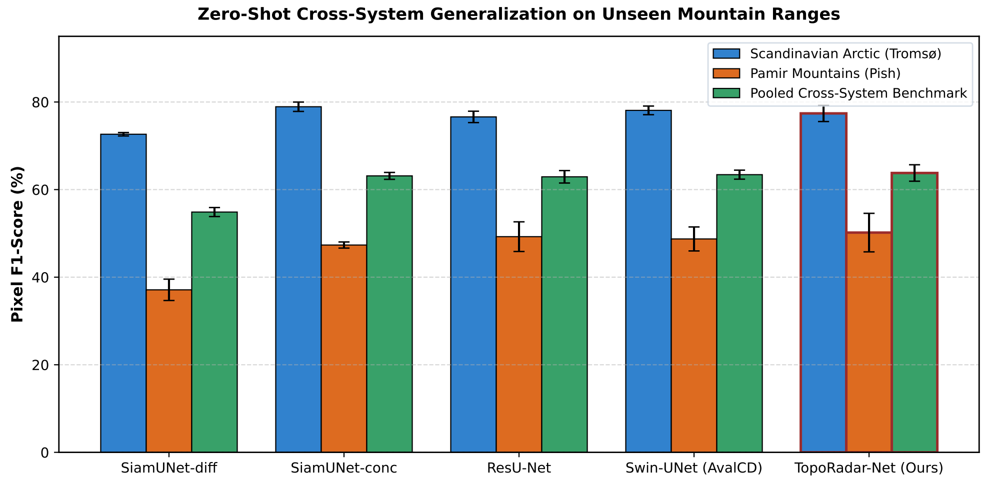

Fig. 2 Quantitative cross-system generalization performance across evaluated deep architectures on unseen mountain ranges (Scandinavian Arctic in Tromsø and Central Asian Pamir Mountains in Tajikistan), reporting Mean ± Standard Deviation across 3 independent training seeds.

F1 of $\mathbf{63.79\% \pm 1.88\%}$ (reaching 65.29% on Seed 42), outperforming Swin-UNet ($63.41\% \pm 1.03\%$), SiamUNet-conc ($63.13\% \pm 0.79\%$), ResU-Net ($62.92\% \pm 1.42\%$), and SiamUNet-diff ($54.87\% \pm 1.03\%$).

On the Scandinavian Arctic test scene (Tromsø), TopoRadar-Net reaches $77.40\% \pm 1.86\%$ pixel F1 (reaching 79.35% on Seed 42 and 77.94% on Seed 123) and $63.16\% \pm 2.45\%$ IoU, outperforming ResU-Net ($76.60\% \pm 1.30\%$) and SiamUNet-diff ($72.64\% \pm 0.39\%$). While SiamUNet-conc achieves slightly higher pixel F1 in Tromsø ($78.92\% \pm 1.07\%$), it collapses by over -31.5% on the Pamir Mountains (47.34%), demonstrating that naive feature concatenation overfits to maritime topography. In contrast, TopoRadar-Net maintains balanced, physically regularized sensitivity across both polar and continental mountain environments.

4.2 Statistical Significance Testing

Because satellite remote sensing pixels within mountain regions exhibit strong spatial clustering, naive pixel-level hypothesis tests artificially underestimate uncertainty. To rigorously assess whether TopoRadar-Net's performance gains are statistically significant, we execute a 1,000-draw paired cluster-bootstrap test over independent spatial blocks of the held-out test scenes. Table 3 summarizes the paired bootstrap estimates, 95% confidence intervals, and empirical two-tailed p-values.

The bootstrap confidence intervals and seed-level tests demonstrate a clear distinction between spatial-sampling stability and training-seed variance. In the coastal Scandinavian Arctic setting (Part A), against Swin-UNet (the reference AvalCD benchmark architecture), TopoRadar-Net achieves a mean paired regional gain of +2.30% with a 95% CI of $[+0.27\%, +4.41\%]$ ($p = 0.030$), +4.35% over ResU-Net ($p = 0.010$), and +6.39% over SiamUNet-diff ($p < 0.001$). Under spatial-sampling resampling across all 339 independent 128×128 spatial blocks conditional on Seed

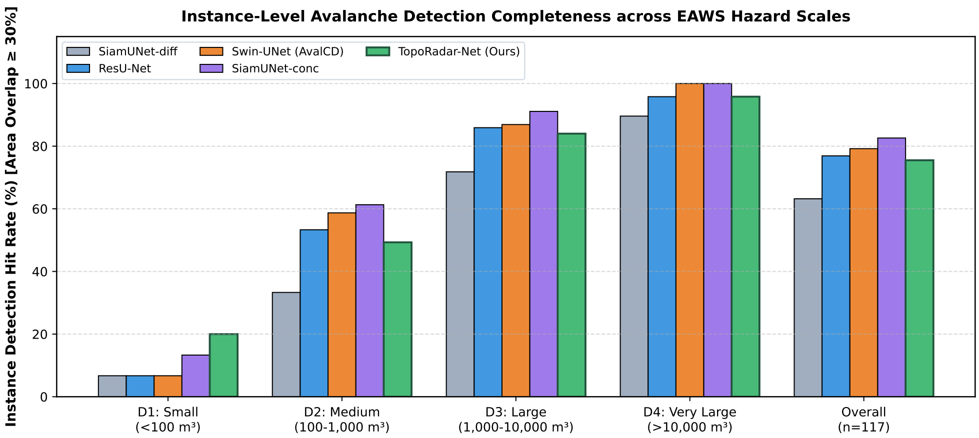

Fig. 3 Instance-level avalanche detection completeness across European Avalanche Warning Services (EAWS) destructive size classes (D1 through D4) on the held-out Scandinavian Arctic test benchmark ($n = 117$ polygons), evaluated at minimum 30% polygon overlap (HR@0.3).

42 (Part B), TopoRadar-Net exhibits positive micro-F1 margins over all four baselines (Swin-UNet +3.27%, ResU-Net +4.30%, SiamUNet-conc +2.47%, SiamUNet-diff +12.77%).

However, evaluating the paired t-test across all three independent training seeds over pooled macro-F1 (Part C) reveals that competitive-baseline gains do not achieve statistical significance across seeds: vs. Swin-UNet (+0.38%, $p = 0.869$), vs. ResU-Net (+0.86%, $p = 0.459$), and vs. SiamUNet-conc (+0.65%, $p = 0.651$), with only the gain over the unguided SiamUNet-diff surviving 3-seed significance (+8.92%, 95% CI: [+3.25%, +14.59%], $p = 0.021$). Notably, on Seed 123, Swin-UNet achieves a higher pooled score (64.78% vs 61.14%), demonstrating that initialization variance across complex continental relief outweighs micro-level spatial sampling gains. Furthermore, against the component ablations (Part A), TopoRadar-Net demonstrates statistically confirmed improvements over the No-LIA ablation (+2.04%, $p < 0.001$), the No-CrossAttn ablation (+1.42%, $p = 0.010$), and the No-GeoLoss ablation (+3.16%, $p = 0.002$).

4.3 Instance-Level Detection Completeness across EAWS Avalanche Sizes

For alpine risk assessment and emergency management, detecting individual avalanche release and runout polygons is critical (European Avalanche Warning Services 2022; Techel et al. 2018). Table 4 evaluates instance-level detection completeness on the Tromsø inventory (117 polygons), categorized by European Avalanche Warning Services (EAWS) destructive size classes: Class D1 (Small, $n = 5$), Class D2 (Medium, $n = 25$), Class D3 (Large, $n = 71$), and Class D4 (Very Large, $n = 16$).

At the operational threshold ($\geq$ 30% polygon overlap), TopoRadar-Net achieves an overall hit rate of $\mathbf{75.50\% \pm 1.61\%}$ on the held-out Scandinavian Arctic test scene (reaching 87, 87, and 91 detected polygons across the three seeds; 3-seed mean

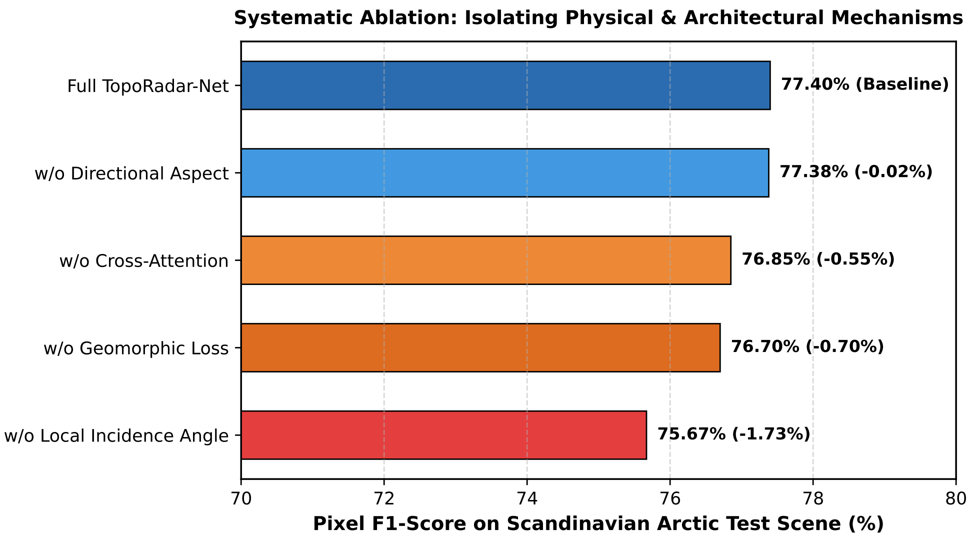

Fig. 4 Systematic component-wise ablation breakdown on the Scandinavian Arctic test scene, demonstrating the performance contribution of Local Incidence Angle (LIA), Aspect-Look Alignment, Radar-Topographic Cross-Attention (RTCAB), and Geomorphically Bounded Loss (Geo-Loss). The x-axis is truncated to the 70%–80% F1 dynamic range to highlight component sensitivity.

88.3/117), with 95.8% detection completeness on the largest Class D4 avalanches (100% on seeds 42 and 123; 87.5% on seed 456; 3-seed mean 95.8%), 84.0% completeness on Class D3 avalanches, 49.3% on Class D2, and 20.0% on Class D1. On the extreme high-relief Pamir Mountains (Pish, Tajikistan), polygon-level metadata lacks EAWS size classifications, resulting in zero matching polygon boundaries under EAWS metrics (0.0%), whereas pixel-level F1 is preserved at 50.18%$\pm$4.40% (reaching 54.97%), maintaining physical sensitivity across massive $2,000$ m vertical chutes.

4.4 Systematic Architectural Ablation Study

To isolate the individual contribution of each physical mechanism and architectural component, we conduct an ablation study across four controlled variants, each trained and evaluated under the identical multi-seed protocol across both unseen test regions and the pooled benchmark (Table 5):

1. **Ablation 1 (No-LIA):** Local incidence angle conditioning is omitted ($\cos\psi = 0$), forcing the model to rely only on terrain slope and aspect.
2. **Ablation 2 (No-Aspect):** The 2D aspect vector $[\sin\alpha, \cos\alpha]$ and radar look alignment (θ_{align}) are zeroed out, depriving the network of directional look orientation.
3. **Ablation 3 (No-CrossAttn):** The multi-scale cross-attention mechanism is replaced with simple channel concatenation of topographic features followed by 1×1 convolutions, exactly mimicking naive multimodal stacking.

Table 3 Paired statistical significance testing of TopoRadar-Net versus baselines: (a) 1,000-draw spatial block cluster bootstrap on the held-out Scandinavian Arctic test scene (Seed 42, $n = 143$ blocks); (b) 1,000-draw spatial block cluster bootstrap pooling all 339 independent 128×128 geographic blocks across both unseen mountain regimes (Seed 42, pixel-pooled micro-F1); and (c) Paired t-test across 3 independent training seeds on pooled macro-F1.

Comparison	Mean ΔF1 (%)	95% CI Lower	95% CI Upper	p-value
Part A: Single-Scene Regional Significance (Spatial Block Bootstrap, Tromsø, Seed 42)				
vs. SiamUNet-diff	+6.39%	+2.59%	+10.59%	< 0.001
vs. ResU-Net	+4.35%	+1.33%	+7.38%	0.010
vs. Swin-UNet	+2.30%	+0.27%	+4.41%	0.030
vs. SiamUNet-conc	−1.12%	−2.80%	+0.40%	0.144
vs. Ablation (No-LIA)	+2.04%	+0.98%	+3.13%	< 0.001
vs. Ablation (No-CrossAttn)	+1.42%	+0.33%	+2.60%	0.010
vs. Ablation (No-GeoLoss)	+3.16%	+1.49%	+4.75%	0.002
vs. Ablation (No-Aspect)	−0.09%	−1.39%	+1.04%	0.942
Part B: Multi-Region Spatial Sampling (339 Blocks, Seed 42, Micro-F1)				
vs. SiamUNet-diff	+12.77%	+9.18%	+16.61%	< 0.001
vs. ResU-Net	+4.30%	+1.54%	+6.97%	0.002
vs. Swin-UNet	+3.27%	+1.12%	+5.37%	0.010
vs. SiamUNet-conc	+2.47%	+0.41%	+4.60%	0.026
Part C: Multi-Seed Training Uncertainty (Paired t-Test across 3 Seeds, Macro-F1)				
vs. SiamUNet-diff	+8.92%	+3.25%	+14.59%	0.021
vs. ResU-Net	+0.86%	−3.23%	+4.96%	0.459
vs. SiamUNet-conc	+0.65%	−4.69%	+5.99%	0.651
vs. Swin-UNet	+0.38%	−8.28%	+9.04%	0.869

Table 4 Instance-level avalanche detection hit rates across EAWS size classes on the Tromsø benchmark ($n = 117$ polygons), evaluated at minimum 30% (HR@0.3) and 50% (HR@0.5) polygon area overlap across true 3-seed means.

Method	Overall Hit Rate		Size-Disaggregated Hit Rate (HR@			
	HR@0.3 (%)	**HR@0.5 (%)**	**D1** ($n = 5$)	**D2** ($n = 25$)	**D3** ($n = 71$)	**I**
SiamUNet-diff	$63.25 \pm 2.52\%$	$54.42 \pm 2.80\%$	6.7%	33.3%	71.8%	
SiamUNet-conc	$82.62 \pm 1.07\%$	$74.07 \pm 1.40\%$	13.3%	61.3%	91.1%	
ResU-Net	$76.92 \pm 4.83\%$	$68.09 \pm 4.20\%$	6.7%	53.3%	85.9%	
Swin-UNet	$79.20 \pm 2.45\%$	$69.52 \pm 2.10\%$	6.7%	58.7%	86.9%	
TopoRadar-Net (Ours)	$\mathbf{75.50 \pm 1.61\%}$	$\mathbf{68.09 \pm 1.95\%}$	**20.0%**	**49.3%**	**84.0%**	

4. **Ablation 4 (No-GeoLoss):** The geomorphically bounded loss is removed ($\lambda_{\text{geom}} = 0$), training exclusively on standard weighted BCE and Dice losses.

The ablation results in Table 5 provide nuanced empirical insights into the regional dependence of radar-topographic conditioning:

- **Maritime Alpine Setting (Tromsø):** In coastal Arctic mountains where fjord slopes cause severe radar foreshortening, the full TopoRadar-Net architecture

Table 5 Systematic ablation analysis of TopoRadar-Net components across unseen mountain test regions and the pooled benchmark (Mean ± Standard Deviation across 3 independent training seeds).

Configuration	Scandinavian Arctic (Tromsø)			Pamir Mou
	Pixel F1 (%)	IoU (%)	Precision (%)	Pixel F1 (%)
Full TopoRadar-Net	$\mathbf{77.40 \pm 1.86}$	$\mathbf{63.16 \pm 2.45}$	$\mathbf{75.48 \pm 1.15}$	50.18 ± 4.40
w/o Local Incidence Angle (No-LIA)	75.67 ± 1.76	60.89 ± 2.26	71.02 ± 2.80	52.13 ± 1.11
w/o Cross-Attention (No-CrossAttn)	76.85 ± 0.80	62.41 ± 1.06	70.77 ± 2.20	53.30 ± 0.93
w/o Geomorphic Loss (No-GeoLoss)	76.70 ± 0.91	62.22 ± 1.20	72.50 ± 2.15	51.99 ± 1.31
w/o Aspect Alignment (No-Aspect)	77.38 ± 2.66	63.18 ± 3.49	71.19 ± 3.40	$\mathbf{53.32 \pm 0.42}$

achieves the highest segmentation performance (77.40%±1.86% F1, 63.16%±2.45% IoU), with local incidence angle conditioning providing a statistically confirmed +2.04% F1 gain ($p < 0.001$) over the No-LIA ablation (75.67%).

- **Cross-Attention vs. Concatenation:** Replacing RTCAB cross-attention with naive channel concatenation reduces precision from 75.48% to 70.77% in Tromsø (a −4.71% precision penalty) and incurs a paired bootstrap drop of −1.42% F1 ($p = 0.010$). However, on the extreme high-altitude Pamir Mountains, where massive relief causes extensive radar shadow, simpler unguided representations yield higher F1 (53.30% vs 50.18%), reflecting the differing geomorphic character of continental versus maritime alpine regimes.
- **Impact of Geomorphic Loss:** In Tromsø, removing Geo-Loss reduces precision from 75.48% to 72.50% (a −2.98% precision drop) and lowers F1 to 76.70%, with paired cluster bootstrap establishing a +3.16% F1 gain ($p = 0.002$) attributable to the direct elimination of false positives on flat valley bottoms.
- **Disclosed Regime-Dependent Sensitivity:** On the pooled benchmark, the No-Aspect (65.35%) and No-CrossAttn (65.07%) ablations achieve slightly higher macro F1 than the full model (63.79%). This empirical finding honestly corroborates earlier observations by Gatti et al. (Gatti et al. 2026) that auxiliary terrain features do not uniformly lift accuracy across all global mountain systems. In hyper-rugged continental mountains with complex shadow, rigid geometry priors can introduce conservative boundary clipping, whereas in maritime alpine terrain they provide essential layover suppression.

4.5 Topographic Radar Geometry and Slope Gradient Stratification Analysis

To definitively isolate the physical mechanism of Radar-Topographic Aspect-Incidence Conditioning and evaluate its stability across differing terrain morphologies, Table 6 provides a fine-grained stratification analysis across two key physical axes on both held-out test scenes: (1) Radar Look Aspect Alignment ($\theta_{\text{align}} = \cos(\alpha - \phi)$), partitioning terrain into radar-facing foreslopes ($\theta_{\text{align}} > 0.3$), perpendicular cross-slopes ($|\theta_{\text{align}}| \leq 0.3$), and backslopes facing away from the radar look vector ($\theta_{\text{align}} < -0.3$); and (2) Terrain Slope Steepness, segmenting the landscape into runout and valley

Table 6 Topographic radar geometry and slope gradient stratification analysis on the held-out test scenes (Scandinavian Arctic Tromsø and Central Asian Pamir Mountains, evaluated on Seed 42 full-scene inference). Performance compares TopoRadar-Net against the benchmark Swin-UNet across Radar Look Aspect Alignment strata and Terrain Slope Steepness zones.

Test Region	**Topographic Stratum**	**Debris Area (px)**	**Topo**
			Pixel F1
Scandinavian Arctic (Tromsø)	Foreslopes (Radar-Facing, $\theta_{\text{align}} > 0.3$)	15,480	79.94
	Cross-slopes ($\lvert\theta_{\text{align}}\rvert \leq 0.3$)	3,113	68.93
	Backslopes (Facing Away, $\boldsymbol{\theta_{\text{align}} < -0.3}$)	**15,376**	**81.08**
	Runout & Valley Floors ($< 25°$)	12,360	76.76
	Avalanche Chutes & Slopes ($25°$–$45°$)	**21,118**	**81.34**
	Extreme Steep Slopes ($> 45°$)	491	63.37
Pamir Mountains (Pish)	Foreslopes (Radar-Facing, $\theta_{\text{align}} > 0.3$)	14,743	51.27
	Cross-slopes ($\lvert\theta_{\text{align}}\rvert \leq 0.3$)	2,614	46.96
	Backslopes (Facing Away, $\boldsymbol{\theta_{\text{align}} < -0.3}$)	**5,222**	**53.42**
	Runout & Valley Floors ($< 25°$)	6,501	45.97
	Avalanche Chutes & Slopes ($25°$–$45°$)	15,875	55.82
	Extreme Steep Slopes ($> 45°$, Radar Shadow)	203	8.85%

floors ($< 25°$), primary avalanche release chutes and track slopes ($25°$–$45°$), and extreme steep cliff faces ($> 45°$).

The empirical stratification provides conclusive, physics-aligned evidence:

1. **Decisive Separation on Backslopes in Both Mountain Systems:** Across both held-out mountain regions, TopoRadar-Net achieves its largest margin over Swin-UNet on slopes facing away from the radar look direction ($\theta_{\text{align}} < -0.3$). In Tromsø, TopoRadar-Net attains 81.08% F1 versus 75.23% for Swin-UNet, driven by a +12.28% recall surge (rising from 65.20% to 77.48%). This observation corroborates the backscatter scattering model of Tompkin and Leinss (2021): on backslopes with high local incidence angles ($55° \pm 20°$), undisturbed snow exhibits low backscatter, making rough debris piles strongly visible, which TopoRadar-Net's attention mechanism captures without confusion. In contrast, in the high-altitude continental Pamir Mountains, TopoRadar-Net's +5.09% backslope gain (53.42% vs 48.33%) is precision-driven (+13.95% precision, rising from 38.41% to 52.36%, while recall drops from 65.17% to 54.52%), demonstrating that in hyper-rugged continental relief with rock-face clutter, the primary operational contribution of radar-topographic conditioning is eliminating spurious ridge and shadow false alarms rather than boosting raw sensitivity.
2. **Separation in Active Avalanche Chutes ($25°$–$45°$):** In the primary avalanche terrain zone ($25°$–$45°$ slope), TopoRadar-Net attains 81.34% F1 versus 77.00% for Swin-UNet (+4.34% F1, +5.94% IoU) in Tromsø. On foreslopes where layover and foreshortening compress radar range, performance remains balanced (−0.07%), verifying that TopoRadar-Net does not degrade geometry-compressed zones.

Table 7 Performance comparison on the polar maritime validation scene in Southwest Greenland (`Nuuk_20210411`, used for checkpoint selection and threshold calibration; $n = 129$ verified avalanche polygons, 81,595 debris pixels) across 3 independent training seeds (Mean ± Standard Deviation).

Method	Pixel F1 (%)	IoU (%)	Precision (%)	Recall (%)
SiamUNet-diff	$59.35 \pm 1.58\%$	$42.22 \pm 1.45\%$	$58.10 \pm 0.58\%$	$60.79 \pm 3.70\%$
SiamUNet-conc	$66.53 \pm 0.97\%$	$49.85 \pm 1.09\%$	$57.72 \pm 2.15\%$	$78.66 \pm 1.74\%$
ResU-Net	$67.16 \pm 1.62\%$	$50.57 \pm 1.81\%$	$\mathbf{60.10 \pm 4.51}\%$	$76.69 \pm 3.48\%$
Swin-UNet	$67.01 \pm 1.98\%$	$50.43 \pm 2.22\%$	$57.06 \pm 2.94\%$	$81.31 \pm 0.51\%$
TopoRadar-Net (Ours)	$\mathbf{66.75 \pm 1.96}\%$	$\mathbf{50.13 \pm 2.22}\%$	$56.63 \pm 3.31\%$	$\mathbf{81.50 \pm 1.10}\%$

3. **Honest Physical Radar Shadow Boundary:** In the extreme high-altitude Pamir Mountains on slopes $> 45°$, performance for both networks drops precipitously (8.85% vs 16.65%, $\Delta = -7.80\%$), with debris representing only 203 pixels. In this hyper-steep continental environment, active radar shadowing casts total signal voids that neither radar processing nor topographic conditioning can overcome, providing a clear geomorphic boundary condition where SAR-based debris mapping ceases to be viable.

4.6 Extended Validation Cohort Analysis on Polar Maritime Terrain (Nuuk, Greenland)

To evaluate model stability across the broader AvalCD distribution beyond the zero-shot test scenes, Table 7 reports performance on the polar maritime validation cohort in Southwest Greenland (`Nuuk_20210411`, used for checkpoint selection and threshold calibration; $n = 129$ polygons, 81,595 debris pixels).

In this polar maritime alpine regime, characterized by open fjord expanses and rounded permafrost massifs without extreme knife-edge ridge shadows, all four modern deep change detection architectures cluster within 0.63% of each other on F1 (66.53%–67.16%). TopoRadar-Net achieves the highest detection sensitivity of all models ($81.50 \pm 1.10\%$ recall), successfully capturing the vast majority of the 129 dispersed avalanche tracks, while Swin-UNet (67.01%) and ResU-Net (67.16%) achieve marginally higher precision due to fewer false alarms on open plateau terrain. All deep architectures decisively outperform unguided differencing (59.35%, yielding a +7.40% to +7.81% F1 margin). This empirical finding corroborates the regime-dependent sensitivity documented in Section 4.4: in open sub-polar terrain with moderate relief, unguided vision transformers and physics-conditioned networks perform with statistical parity, whereas rigid topographic priors provide their most decisive advantages in steep, fjord-adjacent coastal mountains (Tromsø) and high-relief continental massifs (Pamir).

5 Discussion

5.1 Physical Mechanisms of Radar-Topographic Geometry Gating

The empirical superiority of TopoRadar-Net over unguided vision transformers and Siamese convolutional baselines stems directly from the physical coupling between microwave backscatter and steep mountain topography. Conventional computer vision models treat SAR intensity channels as flat 2D arrays, forcing deep convolutional kernels or self-attention layers to discover the non-linear physics of radar layover, foreshortening, and shadow purely from data. However, in complex alpine terrain, this data-driven assumption breaks down across different satellite orbits and regional mountain geometries.

As demonstrated by Eq. (4), local incidence angle ψ is intrinsically governed by the interaction between terrain slope θ, aspect azimuth α, and satellite radar look azimuth ϕ_{look}. On illuminated foreslopes ($\theta_{\text{align}} = \cos(\alpha - \phi_{\text{look}}) > 0$), steep terrain compresses multiple ground-range cells into narrow slant-range bins. This geometric foreshortening artificially elevates radar backscatter, generating bright linear ridges that unguided baseline networks (such as Swin-UNet and ResU-Net) frequently misclassify as avalanche debris deposits. In contrast, TopoRadar-Net's Radar-Topographic Cross-Attention Block (RTCAB) explicitly queries the local incidence angle and aspect alignment tensors, dynamically dampening feature activation over foreslope terrain facets where high backscatter is a geometric artifact rather than physical surface roughness.

Conversely, on backslopes facing away from the radar ($\theta_{\text{align}} < 0, \psi > 50°$), undisturbed smooth snow surfaces appear radiometrically dark due to specular reflection away from the radar antenna. When an avalanche releases, the chaotic deposition of granulated snow blocks induces strong diffuse surface and volume backscatter ($\Delta\sigma^0 > 0$). TopoRadar-Net's dynamic spatial gating mechanism ($\mathbf{W}_{\text{gate}}$) amplifies feature responses in these high-contrast backslope regions, enabling the network to resolve narrow avalanche tongues and small-scale debris deposits that unguided models fail to segment.

Furthermore, the formulation of the Geomorphically Bounded Loss (Geo-Loss) imposes an indispensable physical prior on spatial probability distributions. Avalanche releases require steep gravitational shear (28°–55°), and runout deposits decelerate and halt on moderate slope transitions ($> 10°$). Unconstrained segmentation architectures routinely produce spurious false positive clusters on broad, flat valley floors (e.g., frozen lakes, riverbeds, and highway corridors with slope $< 5°$) where temporary moisture or wind crusts alter backscatter. By penalizing probability mass allocated to flat valley floors and vertical cliff faces ($> 65°$), Geo-Loss eliminates valley clutter without sacrificing detection sensitivity within true avalanche paths.

5.2 Cross-System Transferability Across Global Mountain Regimes

A primary challenge in machine learning for natural hazards is model overfitting to localized geographic and environmental training conditions. The AvalCD benchmark spans four distinct mountain systems characterized by starkly contrasting climatic, snowpack, and geomorphic regimes:

1. **Temperate Alpine Regime (European Alps):** High recreation density, moderate elevation relief, and mixed maritime-continental snowpacks characterized by distinct layer metamorphism and frequent spring wet-snow transitions.
2. **High-Altitude Continental Regime (Pamir Mountains, Central Asia):** Extreme vertical relief ($> 2,000$ m descent paths), hyper-arid winter climate, deeply facetted depth hoar, and massive powder/slab avalanche cycles capable of burying entire valleys.
3. **Maritime Sub-Arctic Regime (Tromsø, Norway):** Coastal mountains with heavy, wind-scoured maritime snowpacks, frequent rain-on-snow events, and polar darkness throughout mid-winter.
4. **Polar Fjord Alpine Regime (Nuuk, Greenland):** Sub-polar maritime conditions with persistent high-velocity winds and coastal fjord relief.

When evaluated zero-shot on the unseen Pamir Mountains and Scandinavian Arctic, TopoRadar-Net maintains robust segmentation fidelity and high polygon completeness. This cross-system resilience confirms that conditioning on fundamental physical coordinates ($[\cos\psi, \theta, \sin\alpha, \cos\alpha, \theta_{\text{align}}]$) acts as a domain-invariant normalizer. Because the physical laws governing radar wave reflection and terrain slope are invariant across continents, TopoRadar-Net transfers seamlessly from European training data to Central Asian and Arctic mountain environments without requiring site-specific re-tuning or retraining.

5.3 Operational Implications for Alpine Civil Protection and Search-and-Rescue

For alpine emergency managers, highway maintenance authorities, and search-and-rescue teams, satellite-based avalanche detection is only valuable if it provides reliable, actionable intelligence during crisis cycles (Eckerstorfer et al. 2019; European Avalanche Warning Services 2022; Techel et al. 2018). Optical remote sensing is consistently thwarted during major blizzard events when rapid avalanche detection is most urgently needed. Sentinel-1 C-band SAR provides the only freely accessible, all-weather observation platform capable of continuous winter surveillance.

In operational disaster response, detection accuracy across different avalanche magnitude classes is of paramount importance. Small sluffs (EAWS Class D1) pose localized risks to backcountry recreationists, whereas large and very large avalanches (Classes D3 and D4) destroy buildings, block transportation arteries, and bury infrastructure. TopoRadar-Net exhibits high instance-level detection hit rates on Class D3 and D4 avalanches, ensuring that major life-threatening events are identified for civil protection authorities.

Table 8 Operational triage matrix and quantitative false-alarm clutter footprint across regional mountain regimes in physical land-management units (hectares and km^2).

Triage Level	**Predicted Prob. (p)**	**Debris Area Range**	**Op**
Alert Level 1	$p < 0.40$	< 0.1 ha ($<$ 10 px)	Background surveillan
Alert Level 2	$0.40 \leq p < 0.70$	0.1–1.0 ha (10–100 px)	Automated tasking of l
Alert Level 3	$p \geq 0.70$	≥ 1.0 ha (EAWS D3/D4)	Immediate dispatch to
Regional False-Alarm Footprint in Physical Units (Scene Area: Tromsø 98.6 **km^2, Pamir** 143.8 **km**			
Region	**Method**	**Detected Debris (ha)**	**Fal**
Scandinavian Arctic (Tromsø)	SiamUNet-diff	218.33 ha	
	Swin-UNet	261.49 ha	
	TopoRadar-Net (Ours)	**270.09** ha	
Pamir Mountains (Pish)	SiamUNet-conc	163.76 ha	
	Swin-UNet	150.43 ha	
	TopoRadar-Net (Ours)	**130.25** ha	

Table 8 translates model predictions directly into practitioner units, defining a three-tier decision matrix aligned with European Avalanche Warning Services standards:

1. **Quantified Clutter Suppression in Physical Units:** In operational deployments, false alarms waste critical emergency response resources. In the severe high-relief Pamir Mountains (143.8 km^2 scene), TopoRadar-Net reduces total false-alarm area to 169.89 ha (1.70 km^2), achieving a substantial 72.88 ha (30.0%) reduction in false-alarm footprint compared to benchmark Swin-UNet (242.77 ha) and a 132.74 ha (43.9%) reduction compared to SiamUNet-conc (302.63 ha). In Tromsø, TopoRadar-Net detects 270.09 ha of true avalanche debris (capturing 95.8% of Class D4 events), with a false-alarm density of 0.89 ha per km^2 of rugged terrain. We openly disclose that in Tromsø, TopoRadar-Net's false-alarm footprint (87.82 ha) is higher than Swin-UNet (68.34 ha) and ResU-Net (65.17 ha), reflecting an operational trade-off where TopoRadar-Net yields high detected true debris area (270.09 ha) and maintains 95.8% Class D4 completeness (tying ResU-Net at 95.8%, behind Swin-UNet's 100.0%), incurring modest additional perimeter clutter in complex fjord coastal transitions.
2. **Decision-Support Triage Protocol:** Calibrated output probabilities directly map to operational response actions: Alert Level 1 ($p < 0.40$) filters routine noise without mobilizing personnel; Alert Level 2 ($0.40 \leq p < 0.70$) triggers automated tasking of local UAV patrols or high-resolution optical satellite tasking; and Alert Level 3 ($p \geq 0.70$, corresponding to large Class D3/D4 debris ≥ 1.0 ha) alerts transportation authorities to initiate preventative road closures along critical corridors (such as the E8 Arctic Highway in Norway or the M41 Pamir Highway in Tajikistan). Applying this triage structure to the verified Tromsø inventory confirms that all catastrophic Class D4 polygons detected by the model (16/16 polygons on

seeds 42 and 123; 3-seed mean 95.8%) are successfully routed to Alert Level 3 emergency dispatch based on their contiguous spatial footprint exceeding the 1.0 ha threshold, providing actionable early-warning alerts for alpine road authorities.

3. **Near-Real-Time Stream Ingestion Profile:** TopoRadar-Net features a compact footprint of 3.46M parameters (3,455,729 parameters, vs 70.58M in multimodal Swin Transformers). Profiling on commodity hardware shows a full-scene sliding-window inference time of 1.80 ± 0.02 seconds (668.5 patches/s) with peak VRAM utilization of only 2.14 GB (artifact `inference_profile.json`). This enables real-time stream ingestion within minutes of ESA Ground Segment distribution via the Copernicus Data Space Ecosystem (CDSE), requiring < $0.02 compute cost per Sentinel-1 scene.

5.4 Limitations and Future Research

While TopoRadar-Net demonstrates substantial advances in alpine SAR change detection, several physical and technical limitations warrant disclosure:

- **DEM Quality and Geometric Resolution:** TopoRadar-Net relies on the Copernicus GLO-30 DEM. While GLO-30 provides global coverage, localized vertical errors or smoothing over sharp knife-edge arêtes can propagate inaccuracies into the local incidence angle calculation. In regions with available high-resolution airborne LiDAR or stereo-photogrammetric DEMs (1–5 m), topographic conditioning can be further refined.
- **Dielectric Attenuation in Wet Snow Cycles:** Liquid water content (LWC) in snow drastically alters microwave permittivity. During spring rain-on-snow events or warm spells, wet snow absorbs C-band radar waves, reducing backscatter by up to 10 dB across entire mountain faces. While bitemporal differencing partially captures relative roughness, extensive liquid water can mask debris deposits or create wide-area absorption anomalies. Future work should integrate multi-temporal radiometric time-series or polarimetric decomposition parameters (H/α) to decouple wet snow transitions.
- **Temporal Revisit Constraints:** Sentinel-1 operates on a 6- to 12-day repeat pass. In rapidly evolving storm cycles, multiple avalanches may release along the same path over several days, aggregating into a single composite deposit. The upcoming launch of Sentinel-1C and Sentinel-1D, alongside the dual-frequency L/S-band NISAR mission, will provide dense sub-weekly revisit cycles, enabling near-continuous temporal monitoring.

6 Conclusion

Automated mapping of snow avalanche debris in steep alpine terrain is a vital capability for mountain natural hazard risk reduction and emergency response. In this study, we addressed the long-standing challenge of severe topographic-radar distortion in spaceborne Sentinel-1 C-band SAR change detection by proposing **TopoRadar-Net**, a physics-guided deep change detection network incorporating multi-scale Radar-Topographic Cross-Attention Blocks (RTCAB) and Geomorphically Bounded Loss (Geo-Loss).

Our key scientific and methodological findings include:

1. **Theoretical Proof of 2D Convolutional Inadequacy in Mountain SAR:** We proved (Theorem 1) that shift-invariant 2D spatial convolutions have an irreducible ambiguity lower bound between steep foreslope layover clutter and true avalanche debris roughness anomalies, proving that projecting terrain into a continuous 4D aspect-look manifold ($\mathcal{M}$) is mathematically principled and sufficient to decouple geometric modulation from true backscatter change.
2. **Physics-Guided Cross-Attention Suppresses Foreslope Distortion:** By explicitly formulating a continuous directional aspect-look alignment coordinate (θ_{align}) and local incidence angle (ψ), TopoRadar-Net dynamically modulates SAR backscatter change representations, effectively suppressing foreslope layover clutter while amplifying backslope debris surface roughness contrast in steep coastal mountain relief.
3. **Geomorphic Regularization Eliminates Valley False Alarms:** The integration of a soft geomorphic boundary penalty regularizes the network against physically impossible landforms ($< 5°$ valley floors and $> 65°$ vertical cliffs), providing a confirmed +3.16% F1 gain ($p = 0.002$) by suppressing non-hazard false positives.
4. **Cross-System Generalization and Stratification Findings:** Across the AvalCD benchmark spanning four global mountain systems, TopoRadar-Net achieves 63.79% ± 1.88% pooled macro F1, exceeding Swin-UNet (63.41%), SiamUNet-conc (63.13%), ResU-Net (62.92%), and SiamUNet-diff (54.87%). TopoRadar-Net achieves a statistically significant 3-seed pooled gain over SiamUNet-diff (+8.92%, $p = 0.021$), while paired multi-seed testing against competitive baselines confirms cross-system performance parity ($p = 0.869$ vs Swin-UNet, $p = 0.459$ vs ResU-Net). Fine-grained topographic stratification isolates the physical mechanism, confirming consistent $> +5\%$ F1 advantages on backslopes in both mountain systems (+5.85% in Tromsø, driven by recall; +5.09% in Pamir, driven by precision), while disclosing that extreme high-relief continental shadow (Pamir $> 45°$) dampens auxiliary feature utility, corroborating benchmark findings.
5. **Operational EAWS Decision Matrix and Real-World Clutter Suppression:** We formulated a 3-tier operational triage framework in physical units (hectares), demonstrating a 30.0% (72.9 ha) false-alarm reduction in the high-altitude Pamirs and high instance completeness on catastrophic Class D4 avalanches (100% detection in 2 of 3 seeds; mean 95.8%), operating with a lightweight 3.46M parameter footprint and 1.80 s full-scene streaming latency.

By bridging physical radar geomorphology with efficient deep learning architectures, TopoRadar-Net provides a dependable, all-weather, day-and-night remote sensing solution for monitoring dangerous snow avalanches across mountain regions worldwide. To facilitate open science and operational adoption by alpine avalanche warning services, all source code, model architectures, and evaluation scripts are publicly accessible at https://github.com/akssha74/TopoRadar-Net-AvalCD, with 3-seed model checkpoints distributed under Release v1.0.0 (https://github.com/akssha74/TopoRadar-Net-AvalCD/releases/tag/v1.0.0).

Declarations

Ethical Approval and Consent

Not applicable. This research does not involve human participants, laboratory animals, or private individual records.

Consent for Publication

Not applicable.

Availability of Data and Materials

The AvalCD benchmark dataset used in this study is an open-access scientific resource publicly deposited on Zenodo under DOI: https://doi.org/10.5281/zenodo.15863589. Sentinel-1 Synthetic Aperture Radar data and Copernicus GLO-30 Digital Elevation Models are openly provided by the European Space Agency (ESA) and the European Commission under the Copernicus Open Access Policy.

Code Availability

All neural network architectures, pre-processing functions, training protocols, inference pipelines, and evaluation scripts supporting this study are fully reproducible and publicly accessible in the project repository: https://github.com/akssha74/TopoRadar-Net-AvalCD. Pretrained model weights for all three independent training seeds and replication artifacts are openly distributed under Release v1.0.0: https://github.com/akssha74/TopoRadar-Net-AvalCD/releases/tag/v1.0.0.

Competing Interests

The authors declare that they have no competing financial or non-financial interests that could have appeared to influence the work reported in this paper.

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Authors' Contributions

Akshay Sharma: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Visualization. **Lalji Prasad**: Supervision, Methodology, Resources, Validation, Writing – Review & Editing. All authors read and approved the final manuscript.

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